Journal of Classification (2019) 36:58–72
https://doi.org/10.1007/s00357-019-09317-5

Comparison of Similarity Measures for Categorical Data in Hierarchical Clustering

Zdeněk Šulc[1] · **Hana Řezanková**[1]

Published online: 2 April 2019
© The Classification Society 2019

Abstract
This paper deals with similarity measures for categorical data in hierarchical clustering, which can deal with variables with more than two categories, and which aspire to replace the simple matching approach standardly used in this area. These similarity measures consider additional characteristics of a dataset, such as a frequency distribution of categories or the number of categories of a given variable. The paper recognizes two main aims. First, to compare and evaluate the selected similarity measures regarding the quality of produced clusters in hierarchical clustering. Second, to propose new similarity measures for nominal variables. All the examined similarity measures are compared regarding the quality of the produced clusters using the mean ranked scores of two internal evaluation coefficients. The analysis is performed on the generated datasets, and thus, it allows determining in which particular situations a certain similarity measure is recommended for use.

Keywords Similarity measures · Nominal variables · Hierarchical cluster analysis · Comparison · Evaluation

1 Introduction

Similarity measures play a key role in many multivariate methods, such as multidimensional scaling, outlier detection, or cluster analysis. They serve to determine (dis)similarity between objects characterized by vectors of values. Commonly used similarity measures for datasets with quantitative variables, e.g., the Euclidean distance or the Manhattan distance (see, e.g., Deza and Deza 2009; Warrens 2016), are well examined and can be found in various software packages. A different situation occurs when categorical variables on the nominal scale (i.e., nominal variables) are used. Then, it is impossible to determine the distances between two categories of nominal variables directly. Nowadays, the most common

✉ Zdeněk Šulc
zdenek.sulc@vse.cz

Hana Řezanková
hana.rezankova@vse.cz

[1] Department of Statistics and Probability, University of Economics, Prague, W. Churchill sq. 4, 130 67 Prague 3, Czech Republic

way is to transform the variables into binary ones (dummies) and use similarity measures for binary variables (see, e.g., Warrens 2008). However, the binary transformation often leads to loss of important relationships among the original variables, (see, e.g., Goodall 1966); thus, it is necessary to consider the use of this approach carefully.

There is a group of similarity measures which is not very well explored and which does not require a binary transformation of the used variables. The basic similarity measure, the well-known simple matching coefficient, only reflects if the categories match or mismatch. More advanced approaches use other characteristics of a dataset to improve similarity determination. The first such measure, which considered relative frequencies of categories, was introduced in Goodall (1966), where it was compared with commonly used measures for binary data at that time, such as the Jaccard coefficient (Jaccard 1912), on a numerical taxonomy dataset. Since then, many other approaches were introduced. In Spärck Jones (1972), a similarity measure based on an inverse document frequency principle was introduced, which assigned higher weights to less frequent matches. There are constantly proposed new similarity measures and algorithms based on this principle (see, e.g., Yi et al. 2016), but their use is usually limited to text mining (including clustering). In Lin (1998), an attempt to create a universal similarity measure was introduced. It was derived from theoretic assumptions about similarity, and its universality of use was demonstrated in different domains of the text mining. In Eskin et al. (2002), the proposed similarity measure used the number of categories of a variable for similarity determination. The measure was originally introduced for unsupervised anomaly detection using the *k*-means clustering. It was evaluated on two intrusion datasets, and it showed very promising results. There are other interesting approaches for similarity measures for nominal data. For instance, in Morlini and Zani (2012), a similarity measure for nominal variables which uses a binary transformation was developed. It is an information-based similarity measure which was evaluated in clustering on both the real-world and generated datasets.

The approaches presented in the previous paragraph were evaluated mainly in papers where they were introduced, often with outstanding results. There are only a few papers, where these measures were independently compared and evaluated, for instance, in Boriah et al. (2008) and Chandola et al. (2009), where the selected similarity measures for nominal data were evaluated in a domain of outlier detection. It remains unclear how these measures perform in cluster analysis. The aim of the current paper is to explore which similarity measures work best in the context of hierarchical cluster analysis (HCA), particularly in clustering of objects. Although the objects can be also clustered in various ways, e.g., using the *k*-modes algorithm (Chatuverdi et al. 2001; Huang 1998) or the *k*-representatives algorithm (San et al. 2004), for the similarity measures comparison, we prefer methods of hierarchical cluster analysis. These methods are independent on the order of objects and the linkage algorithms and the assignment of objects into clusters are unambiguous.

There are also approaches of nominal variable clustering and clustering of categories. For these purposes, different similarity measures are usually applied. For nominal variable clustering, association measures between two variables are standardly used. These metrics are well-known; their application in cluster analysis has been already described in Anderberg (1973). Some of the association measures are based on the principle of independence in the contingency table defined by the Pearson chi-square statistic. This principle is also applied for clustering of categories (see Řezanková 2009).

The aim of this paper is to compare similarity measures suitable only for clustering of objects. The final cluster assignment obtained by HCA is influenced not only by the used similarity measure but also by the choice of a linkage method. To investigate this influence, HCA with three commonly used linkage methods is examined in this paper, namely,

single, complete, and average linkage methods. They differ in the way of treating the dissimilarity between two clusters. The single linkage method defines it as the dissimilarity between two closest objects from two different clusters, the complete linkage method treats it as the dissimilarity between two farthest objects from different clusters, and the average linkage method takes the average pairwise dissimilarity between objects in two different clusters.

In the current paper, 11 already proposed similarity measures are compared and evaluated concerning their clustering performance in HCA on categorical datasets. Moreover, two new similarity measures for nominal data are introduced in this paper. The comparison of all 13 similarity measures is performed using the single, complete, and average linkage methods on 60 generated datasets with various numbers of variables and categories. The produced clusters are evaluated by two internal evaluation coefficients using their mean ranked scores. The aim is to determine which similarity measures perform well and under what conditions.

The rest of the paper is organized as follows. Section 2 introduces all similarity measures evaluated in this paper, and Section 3 presents three methods of HCA, and Section 4 describes the used evaluation coefficients. Section 5 contains the experimental part of the paper. Finally, Section 6 presents a summary of the results and a discussion.

2 Similarity Measures for Categorical Data

In this section, the similarity measures examined in this paper are divided into three groups. The first one contains previously introduced similarity measures for nominal data, the second one, a similarity measure based on binary-coded variables, and the third one, the newly proposed similarity measures for nominal data.

2.1 Similarity Measures for Nominal Data

In this paper, similarity measures for nominal data are defined as those which are determined to deal with nominal variables with more than two categories and do not need a dummy transformation. All these similarity measures are applied directly to the data matrix $\mathbf{X} = [x_{ic}]$, where $i = 1, 2, \ldots, n$ (n is the total number of objects); $c = 1, 2, \ldots, m$ (m is the total number of variables). The number of categories of the cth variable is denoted as K_c, an absolute frequency as f, and a relative frequency as p.

When determining the similarity between the objects $\mathbf{x}_i = [x_{i1}, x_{i2}, \ldots, x_{im}]$ and $\mathbf{x}_j = [x_{j1}, x_{j2}, \ldots, x_{jm}]$, this kind of measures is usually computed in two steps. In the first step, similarities between pairs of categories $S_c\left(x_{ic}, x_{jc}\right)$ are computed separately for each variable. The similarity computation differs for matching and non-matching categories (see Table 1), where the column $x_{ic} = x_{jc}$ presents similarity computation for matches, and the column $x_{ic} \neq x_{jc}$ for mismatches. The measures G1, G2, and LIN1 use a subset of the variable $\mathbf{x}_c$ denoted as $\mathbf{q}_c$ for their computation. For the G1 measure, it is defined as $\mathbf{q}_c \subseteq \mathbf{x}_c : \forall q,\, p\,(q) \leq p\,(x_{ic})$, for the G2 measure as $\mathbf{q}_c \subseteq \mathbf{x}_c : \forall q,\, p\,(q) \geq p\,(x_{ic})$, and for the LIN1 measure as $\mathbf{q}_c \subseteq \mathbf{x}_c : \forall q,\, p\,(x_{ic}) \leq p\,(q) \leq p\left(x_{jc}\right)$.

In the second step, the total similarity $S\left(\mathbf{x}_i, \mathbf{x}_j\right)$ between the objects is determined. For the majority of the examined similarity measures (see Table 1), it is calculated as the arithmetic mean over all variables

$$S\left(\mathbf{x}_i, \mathbf{x}_j\right) = \frac{\sum_{c=1}^{m} S_c\left(x_{ic}, x_{jc}\right)}{m}. \tag{1}$$

Table 1 An overview of the used similarity measures for nominal data

Measure	$x_{ic} = x_{jc}$	$x_{ic} \neq x_{jc}$	$S\left(\mathbf{x}_i, \mathbf{x}_j\right)$	$D\left(\mathbf{x}_i, \mathbf{x}_j\right)$
ES	1	$\frac{K_c^2}{K_c^2+2}$	Eq. (1)	Eq. (4)
G1	$1 - \sum_{q \in Q} p^2(q)$	0	Eq. (1)	Eq. (3)
G2	$1 - \sum_{q \in Q} p^2(q)$	0	Eq. (1)	Eq. (3)
G3	$1 - p^2(x_{ic})$	0	Eq. (1)	Eq. (3)
G4	$p^2(x_{ic})$	0	Eq. (1)	Eq. (3)
IOF	1	$\frac{1}{1+\ln f(x_{ic}) \cdot \ln f(x_{jc})}$	Eq. (1)	Eq. (4)
LIN	$2 \ln p(x_{ic})$	$2 \ln \left(p(x_{ic}) + p\left(x_{jc}\right)\right)$	Eq. (2)	Eq. (4)
LIN1	$\sum_{q \in Q} \ln p(q)$	$2 \ln \sum_{q \in Q} p(q)$	Eq. (2)	Eq. (4)
OF	1	$\frac{1}{1+\frac{n}{\ln f(x_{ic})} \cdot \frac{n}{\ln f\left(x_{jc}\right)}}$	Eq. (1)	Eq. (4)
SM	1	0	Eq. (1)	Eq. (3)

Source: Boriah et al. (2008)

For some of the similarity measures, the total similarity is expressed as

$$S\left(\mathbf{x}_i, \mathbf{x}_j\right) = \frac{\sum_{c=1}^{m} S_c\left(x_{ic}, x_{jc}\right)}{\sum_{c=1}^{m} \left(\ln p(x_{ic}) + \ln p\left(x_{jc}\right)\right)}. \tag{2}$$

In order to compute a proximity matrix, which is required by the majority of software solutions, it is necessary to compute dissimilarities $D\left(\mathbf{x}_i, \mathbf{x}_j\right)$ between all pairs of objects, which can be simply obtained from similarities. The dissimilarities are calculated as

$$D\left(\mathbf{x}_i, \mathbf{x}_j\right) = 1 - S\left(\mathbf{x}_i, \mathbf{x}_j\right) \tag{3}$$

for the similarity measures, which take values from zero to one (see Table 1), or as

$$D\left(\mathbf{x}_i, \mathbf{x}_j\right) = \frac{1}{S\left(\mathbf{x}_i, \mathbf{x}_j\right)} - 1 \tag{4}$$

for the measures, which can exceed the value one.

This paper compares ten selected similarity measures for nominal data. The most well-known coefficient in this paper is the measure SM (simple matching), which was introduced in Sokal and Michener (1958). It only recognizes if two categories match or not. Thus, it neglects important characteristics of a dataset, such as the number of categories or the absolute frequencies of categories, which could be used for a better similarity determination. Nevertheless, the SM coefficient is still the most used similarity measure for nominal data. It is used as a part of other similarity measures as well, for instance, in the Gower similarity coefficient (Gower 1971), which serves for similarity determination between objects characterized by the mixed-type variables.

The measure ES was originally proposed in Eskin et al. (2002). It uses the *number of categories* K_c of the cth variable to determine the similarity between two categories. It was constructed to assign higher weights to mismatches for variables with more categories.

Thus, a mismatch in a variable where only two categories occur leads to a much lower similarity than a mismatch in a variable where all categories are unique. In the case of a match, the value one is assigned.

Measures OF (occurrence frequency) and IOF (inverse occurrence frequency) use the *absolute frequencies of the observed categories* to achieve a more precise similarity definition between two categories in the case of mismatches. Originally, these measures were introduced in the field of information retrieval, where they served to determine a relative number of documents containing a specific word (Spärck Jones 1972). The original measures were designed to deal only with binary variables; later, they were adjusted to deal with nominal variables as well. The measures differ by their weight system. Whereas OF assigns higher weights to more frequent mismatches, IOF assigns higher weights to less frequent mismatches. Thus, OF attains its minimum, when both the mismatched categories occur only once in a given variable, and it attains the maximum when there are only two categories in this variable, each of them occurring exactly in half of the cases. By the IOF measure, the situation is exactly the opposite.

Measures G3 and G4, introduced in Boriah et al. (2008) as derivatives of the original Goodall's measure (Goodall 1966), and the LIN measure, introduced in Lin (1998), incorporate the *relative frequencies of observed categories* for similarity determination. Measures G3 and G4 change their values when the categories match; otherwise, they take on a value zero. Overall, they do not exceed the interval from zero to one. The G3 measure assigns higher weights if matching categories are infrequent, and the G4 measure assigns higher weights if matching values are frequent. The LIN measure (Lin 1998) represents an information-theoretic definition of similarity based on relative frequencies. It was derived from theoretic assumptions about similarity with the aim to emphasize a universality of its use. It assigns higher weights to more frequent categories in the case of a match, and lower weights to less frequent categories in the case of a mismatch.

Measures G1, G2, and LIN1, introduced in Boriah et al. (2008) as derivatives of the original Goodall's and Lin's measures (Goodall 1966; Lin 1998), use *relative frequencies of selected categories* to express the similarity between two categories. The G1 measure uses not only a relative frequency of the observed category but also all the relative frequencies which are lower than the observed one. Similarly, the G2 measure considers all the relative frequencies which are higher or equal to the observed one. Both the G1 and G2 measures assign higher weights to infrequent categories in the case of a match. The LIN1 measure contains a complex weight system. In Boriah et al. (2008), it is described in the following way: "It gives lower weight to mismatches if either of the mismatching values is very frequent, or if there are several values that have a frequency in between those of mismatching values. Higher weight is given when there are mismatches on infrequent values, and there are a few other infrequent values. For matches, lower weight is given for matches on frequent categories or matches on values that have many other values of the same frequency. Higher weight is given to matches on rare values."

2.2 A Similarity Measure for Binary-Coded Variables

The MZ measure, introduced in Morlini and Zani (2012) under the name S2, is the only similarity measure examined in this paper which requires binary-coded variables (dummy coding). Thus, for its use, the original nominal variables must be transformed first. This similarity measure was included for a comparison, since it was originally intended for nominal variables by its authors. On the contrary to the similarity measures for binary data based on 2×2 contingency table (see, e.g., Warrens 2008), which are often highly correlated

(see Todeschini et al. 2012), it provides substantially different outcomes. Because of these reasons, it was included for the comparison as a special case. The MZ measure expresses similarity between the binary-coded objects $\mathbf{x}_i$ and $\mathbf{x}_j$ as

$$S\left(\mathbf{x}_i, \mathbf{x}_j\right) = \frac{\sum_{c=1}^{m} \sum_{u=1}^{K_c} \tau\left(i, j\right) \ln\left(\frac{1}{f_{cu}^2}\right)}{\sum_{c=1}^{m} \sum_{u=1}^{K_c} \tau\left(i, j\right) \ln\left(\frac{1}{f_{cu}^2}\right) + \sum_{c=1}^{m} \phi\left(i, j\right) \sum_{u=1}^{K_c} f_{cu} \ln\left(\frac{1}{f_{cu}^2}\right)}, \tag{5}$$

where $u = 1, 2, \ldots, K_c$ is an index of the uth dummy variable of the cth original variable, f_{cu} is an absolute frequency of the uth category of the original cth variable. In order to determine logical operators in Eq. (5), the following rules are set:

$$\begin{aligned}
\tau\left(i, j\right) &= 1 \quad \text{if} \quad x_{icu} = 1 \cap x_{jcu} = 1, \\
&= 0 \quad \text{otherwise}, \\
\phi(i, j) &= 1 \quad \text{if} \quad x_{icu} = 1 \cap x_{jct} = 1 \quad \text{and} \quad x_{ict} = 0 \cap x_{jcu} = 0 \quad \text{for} \quad u \neq t, \\
&= 0 \quad \text{otherwise}.
\end{aligned}$$

The measure is based on Shannon's information theory (Shannon 2001), and it takes values from zero to one. The value zero occurs, when the objects $\mathbf{x}_i$ and $\mathbf{x}_j$ share no positive match and the value one, when the objects are identical.

2.3 Newly Proposed Similarity Measures

In this paper, two new variability-based similarity measures for nominal data are proposed. The *Variable Entropy* (VE) and *Variable Mutability* (VM) measures treat similarity between categories x_{ic} and x_{jc} by the cth variable according to the within-cluster variability of this particular variable. The idea is very simple. Let us assume two different variables, one with large variability, i.e., with approximately evenly distributed categories, and the other one with small variability, where one category is dominant, and the rest of categories is sparsely represented. The new similarity measures give more weight to a match of two categories in the variable with large variability because it is rarer than a match in the low-variability variable. For individual variables variability definition, the VE measure uses the entropy, and the VM measure uses the nominal variance, also known as the mutability or the Gini coefficient. Both of these variability measures use *relative frequencies of all categories*. The VE measure expresses similarity between two categories using the formula

$$S\left(x_{ic}, x_{jc}\right) = \begin{cases} -\frac{1}{\ln K_c} \sum_{u=1}^{K_c} p_u \ln p_u & \text{if} \quad x_{ic} = x_{jc} \\ 0 & \text{otherwise} \end{cases}, \tag{6}$$

where p_u denotes the relative frequency of the uth category ($u = 1, \ldots, K_c$) with $p_u \ln(p_u) = 0$ if $p_u = 0$ to avoid indeterminacy.

For the VM measure, the formula

$$S\left(x_{ic}, x_{jc}\right) = \begin{cases} \frac{K_c}{K_c - 1} \left(1 - \sum_{u=1}^{K_c} p_u^2\right) & \text{if} \quad x_{ic} = x_{jc} \\ 0 & \text{otherwise} \end{cases} \tag{7}$$

is used. The similarity between two objects $S\left(\mathbf{x}_i, \mathbf{x}_j\right)$ is computed analogically to Eq. (1), and the dissimilarity $D\left(\mathbf{x}_i, \mathbf{x}_j\right)$ using Eq. (3).

Both measures were developed with the aim to be simple, not computationally demanding, and theory-based. They take on values from zero to one in the case of a match and zero otherwise. In the case of a match, the zero value can be attained if a given variable takes on only one category for all cases in a dataset. In this case, the variable becomes redundant for clustering, and its zero value is thus fully justified. The value one occurs in matches if all categories by a given variable are equally distributed. When determining the overall similarity between the objects $\mathbf{x}_i$ and $\mathbf{x}_j$, both the similarity measures can take on values from the range zero to one as well. The zero similarity is obtained if there is no match over all variables (or they contain only one category as mentioned before). The value one is achieved if there are matches over all variables and all of them containing the maximal possible variability. In this special case, the newly proposed measures provide the same results as the measures G1 and G2. Furthermore, in this particular case, the relation with the VM measure can be expressed as $\text{VM} = K_c/(K_c - 1) \cdot \text{G1}$, and $\text{VM} = K_c/(K_c - 1) \cdot \text{G2}$.

3 Methods of Hierarchical Clustering

Hierarchical cluster analysis (see, e.g., Everitt et al. 2011; Hennig et al. 2015) is based on either agglomerative or divisive clustering process. Usually, the agglomerative process is used, which works in the following way. In the first step, each object in a dataset is considered as a cluster on its own. Based on a proximity matrix containing dissimilarities for all pairs of clusters, a pair of objects with the lowest dissimilarity is merged into a new cluster. Then, a new proximity matrix between all clusters is calculated, and the algorithm repeats until there is only one cluster left. Using a certain between-cluster distance, a hierarchical structure is created, which can be cut off at any point, and thus, multiple cluster solutions with different numbers of clusters can be obtained. The hierarchical structure can also be displayed in the graphical form of a dendrogram.

An important step in cluster analysis is a choice of a linkage method which sets rules for between-cluster distances definition. Among commonly used linkage methods, there are some that are limited to quantitative data, e.g., Ward's method, which is limited to the squared Euclidean distance, or the centroid method, which cannot be applied to categorical data due to an undefined concept of centroids in this type of data. Although there are ways to deal with these limitations (see, e.g., Strauss and von Maltitz 2017; Chen and Guo 2014), for a comparison of the examined similarity measures in this paper, three commonly used linkage methods are used, namely, the single, complete, and average linkage methods.

The *single linkage method* (SLM) defines the dissimilarity between clusters C_g and C_h as the distance between two closest objects of two different clusters. This linkage is often associated with a problem called the *chaining phenomenon*, in which clusters are merged based on their closest elements, even though, some of their elements are very distant (see, e.g., Everitt et al. 2011). However, this linkage method performs better than other linkages when the clusters are not spherical or elliptical in shape. The *complete linkage method* (CLM) considers a dissimilarity between two clusters $D(C_g, C_h)$ as the dissimilarity between two farthest objects from these clusters. This between-cluster distance usually provides compact clusters with approximately equal diameters. However, it is sensitive to outliers. The *average linkage method* (ALM) takes the average pairwise dissimilarity between objects in two different clusters. It is a robust method, which is considered a compromise between the single and the complete linkage methods (see, e.g., Yim and Ramdeen 2015). An overview of the used linkage method formulas occurs in Table 2, where n_g and n_h are the numbers of objects in clusters C_g and C_h.

Table 2 An overview of the linkage methods

Linkage method	$D(C_g, C_h)$
Single	$\min_{\mathbf{x}_i \in C_g, \mathbf{x}_j \in C_h} D(\mathbf{x}_i, \mathbf{x}_j)$
Complete	$\max_{\mathbf{x}_i \in C_g, \mathbf{x}_j \in C_h} D(\mathbf{x}_i, \mathbf{x}_j)$
Average	$\frac{1}{n_g n_h} \sum_{\mathbf{x}_i \in C_g} \sum_{\mathbf{x}_j \in C_h} D(\mathbf{x}_i, \mathbf{x}_j)$

4 Cluster Evaluation

In this paper, the examined similarity measures are compared and evaluated based on their clustering performance in HCA. Thus, the measures are compared indirectly using the obtained clusters quality, which is measured using two internal evaluation coefficients based on the within-cluster variability. The *pseudo F coefficient based on the mutability* (PSFM) and *pseudo F coefficient based on the entropy* (PSFE) were introduced in Řezanková et al. (2011). Both the coefficients are based on the total within-cluster variability, either expressed by the entropy or by the mutability (the Gini coefficient) which represent two common ways of measuring the variability for nominal variables. Since the within-cluster variability always decreases with an increasing number of clusters, the coefficients are constructed in a way that they penalize an increasing number of clusters. Then, their highest value across several cluster solutions indicates the optimal number of clusters in terms of the used coefficient.

The PSFE coefficient can be computed using the formula

$$\mathrm{PSFE}\,(k) \;=\; \frac{(n-k)\,(\mathrm{WCE}\,(1) - \mathrm{WCE}\,(k))}{(k-1)\,\mathrm{WCE}\,(k)}, \tag{8}$$

where WCE(1) is the variability in the whole dataset with n objects, and WCE(k) the within-cluster variability in the k-cluster solution, which is computed as

$$\mathrm{WCE}\,(k) = \sum_{g=1}^{k} \frac{n_g}{n \cdot m} \sum_{c=1}^{m} \left(- \sum_{u=1}^{K_c} \left(\frac{n_{gcu}}{n_g} \ln \frac{n_{gcu}}{n_g} \right) \right),$$

where n_g is the number of objects in the gth cluster ($g = 1, \ldots, k$), n_{gcu} is the number of objects in the gth cluster by the cth variable with the uth category ($u = 1, \ldots, K_c$).

The PSFM coefficient is computed as

$$\mathrm{PSFM}\,(k) \;=\; \frac{(n-k)\,(\mathrm{WCM}\,(1) - \mathrm{WCM}\,(k))}{(k-1)\,\mathrm{WCM}\,(k)}, \tag{9}$$

where WCM(1) is the variability in the whole dataset with n objects, and WCM(k) the within-cluster variability in the k-cluster solution, which is computed as

$$\mathrm{WCM}\,(k) = \sum_{g=1}^{k} \frac{n_g}{n \cdot m} \sum_{c=1}^{m} \left(1 - \sum_{u=1}^{K_c} \left(\frac{n_{gcu}}{n_g} \right)^2 \right).$$

When dealing with a large number of datasets, there is a need to process and interpret such an amount of data in a synoptic way. In this paper, clusters are evaluated based on the mean ranked scores of the internal evaluation coefficients PSFE and PSFM. Values of these coefficients can be compared not only with their values in different cluster solutions of a certain similarity measure but also with their values in a certain cluster solution for different similarity measures. To compare these values across different datasets, the use of

rank scores, instead of original values of the coefficients, is necessary, because the original values depend on a dataset size and also on the initial variability of a whole dataset WCE resp. WCM (see Eqs. (8) and (9)). The use of ranked scores is one way to make the outcomes of the PSFE and PSFM coefficients comparable.

The evaluation process can be described in two steps. In the first one, using the PSFE (resp. PSFM) coefficient, clusters produced by HCA with all the examined similarity measures in the first dataset are evaluated. The outcome scores are then ranked in a way that the lowest rank is assigned to the highest value of the coefficient. In the same manner, the rank scores for the other datasets are obtained. In the second step, the ranked scores are averaged over the number of replications and other characteristics that are not of interest in the given analysis. For example, to get the mean ranked scores broken down by the number of variables, it is necessary to average the ranked scores over all other characteristics which are not of interest (the number of replications, dataset difficulty, and the number of clusters). The resulting mean ranked scores and their standard deviations are considered as the main output that can be displayed in the form of an easily interpretable table. The lower the mean ranked score of a similarity measure, the better is its clustering performance. The lower value of the standard deviation, the more stable is the clustering performance of a given similarity measure.

5 Experiment

In order to compare and evaluate the clustering performance of the examined similarity measures, analysis on generated datasets was performed. The aim of the analysis is to determine in which particular situations a certain similarity measure creates good-quality clusters and when not. There are many factors which can influence the clustering performance of a similarity measure, e.g., the number of variables or the number of categories. To perform such analysis, use of generated datasets with desired properties is necessary.

Using the adjusted *genRandomClust* function in the *clusterGeneration* R package (see Qiu and Joe 2015), 60 categorical datasets were generated for the experiment. The generation was performed in two steps. In the first one, numeric datasets with the desired numbers of variables and clusters following the multivariate normal distribution were generated, (see Qiu and Joe 2016). In the second step, the numeric variables were divided into the desired number of categories. The resulting categories are in fact ordinal, but the examined similarity measures for nominal data are not influenced by order of categories, so it does not affect the intended analysis.

Apart from the total clustering performance of the examined similarity measures, the influence of the number of variables, the number of categories, and the linkage methods on the quality of clusters was studied. Datasets with four, six, eight, and ten variables, and three types of a difficulty structure, easy (E), medium (M), and hard (H), based on the number of

Table 3 Numbers of categories for 12 generated dataset types

Variables	E	M	H
4	(2,4,3,2)	(2,5,3,5)	(6,10,8,6)
6	(2,4,3,2,4,2)	(2,5,3,5,6,3)	(6,10,8,6,8,10)
8	(2,4,3,2,4,2,2,4)	(2,5,3,5,6,3,2,5)	(6,10,8,6,8,10,6,10)
10	(2,4,3,2,4,2,2,4,3,2)	(2,5,3,5,6,3,2,5,3,5)	(6,10,8,6,8,10,6,10,8,6)

Table 4 Evaluation of clustering of all generated datasets

Index	char.	ES	G1	G2	G3	G4	IOF	LIN	LIN1	MZ	OF	SM	VE	VM
PSFE	M	*5.2*	6.9	7.9	7.4	9.6	6.6	*6.0*	7.3	7.8	6.9	7.4	*5.8*	6.1
	SD	2.5	2.5	2.5	2.6	1.8	2.6	2.8	2.7	2.8	2.6	2.9	2.7	2.7
PSFM	M	*4.0*	7.5	7.9	7.8	8.6	*5.2*	6.3	8.7	8.7	8.0	6.9	*5.4*	6.0
	SD	2.3	2.4	2.2	2.4	1.6	2.5	2.6	2.6	2.4	2.5	2.8	2.5	2.7

categories, were used. Together, that makes 12 combinations of dataset types. An overview of the numbers of categories in each combination is shown in Table 3. To ensure robustness of the obtained results, each combination was replicated five times. Moreover, to eliminate other possible influences, such as the minimal between-cluster distance and the number of clusters, all datasets were generated with the same minimum distance of clusters (parameter $sepVal = 0.3$, which represents partly overlapping clusters in the original numeric dataset) and four clusters (which represent a reasonable compromise between the high number of clusters that is difficult to interpret and two clusters that are often produced by bad-quality clustering). The numbers of objects in generated datasets were not firmly set; they varied from 300 to 700 cases, since the individual cluster sizes were randomly chosen between the values 100 and 200.

For each generated dataset, a set of proximity matrices was computed using the similarity measures presented in Section 2. Next, HCAs with the single, complete, and average linkage methods for the two- to the six-cluster solutions were performed on each proximity matrix. Each of these solutions was evaluated using the evaluation process presented in Section 4. The experiment was performed using the *nomclust* package for R, which was introduced in Šulc and Řezanková (2015). The examined similarity measures were evaluated from the point of view of the number of variables, the difficulty structure of the datasets, and also regarding the used linkage method. In the second part of the analysis, combinations of similarity measures and linkage methods were also evaluated.

Table 4 presents the total results over all types of datasets and linkage methods. There are two characteristics (char.) for each examined similarity measure; mean ranked scores (M) and their standard deviations (SD). Since 13 similarity measures are evaluated in the analysis, the average mean ranked score is seven. Thus, measures evaluated by a lower score than seven provide better clusters than the average and vice versa. The top three scores for the PSFE and the PSFM coefficients are italicized. Between both these coefficients, there

Table 5 Evaluation of clustering on generated datasets broken by the number of variables

Index	var.	ES	G1	G2	G3	G4	IOF	LIN	LIN1	MZ	OF	SM	VE	VM
PSFE	4	*4.6*	7.5	8.8	8.0	10.8	5.2	7.1	7.3	7.1	8.4	7.6	*4.1*	*4.6*
	6	*4.2*	7.0	8.3	7.7	9.2	6.4	6.3	7.5	8.2	7.4	7.8	*5.3*	*5.6*
	8	*5.5*	6.7	7.1	6.8	9.3	7.3	*5.6*	7.0	8.3	*6.4*	7.0	7.1	7.0
	10	6.7	*6.4*	7.4	7.1	9.1	7.6	*5.1*	7.4	7.5	*5.5*	7.1	6.8	7.4
PSFM	4	*3.3*	8.3	8.8	8.4	10.1	*3.9*	7.1	8.7	8.1	9.6	6.7	*3.6*	4.5
	6	*3.0*	7.6	8.2	8.2	8.5	*4.7*	6.4	9.1	9.0	8.4	7.3	*4.9*	5.6
	8	*4.4*	7.1	7.3	7.4	8.1	*6.0*	*6.0*	8.2	9.2	7.2	6.6	6.6	6.8
	10	*5.3*	7.1	7.1	7.4	7.8	*6.4*	*5.5*	8.8	8.6	6.8	6.8	*6.4*	7.0

Table 6 Evaluation of clustering on generated datasets broken down by the difficulty structure

Index	diff.	ES	G1	G2	G3	G4	IOF	LIN	LIN1	MZ	OF	SM	VE	VM
PSFE	E	*5.4*	*6.1*	8.2	7.0	10.6	6.3	6.2	7.6	6.8	7.2	7.3	6.2	*6.0*
	M	*4.7*	6.9	8.3	7.3	10.1	6.2	*5.9*	7.8	8.0	6.5	7.3	*5.8*	6.2
	H	*5.6*	7.6	7.3	7.8	8.2	7.3	*5.9*	6.5	8.5	6.9	7.6	*5.6*	6.2
PSFM	E	*4.0*	6.8	8.4	7.3	10.0	*5.2*	6.0	8.6	8.4	8.3	6.6	*5.6*	*5.6*
	M	*3.2*	7.7	8.4	8.0	9.2	*4.9*	6.1	9.2	8.8	7.8	6.7	*5.1*	5.8
	H	*4.7*	8.1	6.7	8.2	6.6	*5.6*	6.7	8.3	9.0	8.0	7.2	*5.4*	6.5

are not substantial differences regarding the mean ranked scores. However, when they are compared in detail, the PSFE coefficient systematically overestimates the performance of measures LIN1 and MZ, whereas the PSFM coefficient overrates measures ES and IOF.

The lowest mean ranked scores (the best results) with relatively low standard deviations attained the ES measure according to both the evaluation coefficients. The second best clusters were provided by the newly introduced similarity measure VE, which were closely followed by measures IOF, VM, and LIN. The scores of the remaining measures were below the average.

The next part of the analysis aims to evaluate the clustering performance of the examined similarity measures by different numbers of variables (var.). The results are displayed in Table 5. The ES measure performed very well when its performance worsened only slightly by the increasing number of variables. The measures VE, IOF, and VM provided very good clusters, especially, by lower numbers of variables. With an increasing number of variables their performance worsened more substantially, but still, it was around the average. On the contrary, the mean ranked scores of the measures LIN and OF and all the Goodall's measures get better if the number of variables increases. However, from these measures, only LIN performs fairly well on datasets with lower numbers of variables.

The next goal is to determine if the examined similarity measures are influenced by the dataset difficulty structure (diff.), which is expressed by numbers of categories of variables in datasets. The output broken by a difficulty structure is displayed in Table 6. Since the mean ranked scores over three different difficulty structures do not differ very much by the majority of the similarity measures, it seems that the numbers of categories play a minor role in clustering performance of these measures. There are three measures, which prefer a more difficult structure of a dataset, namely, G4, LIN1, and G2 measures. Unfortunately, their clusters based on datasets with fewer categories are very poor.

Table 7 Evaluation of clustering on generated datasets broken down by the linkage method

Index	link.	ES	G1	G2	G3	G4	IOF	LIN	LIN1	MZ	OF	SM	VE	VM
PSFE	SLM	*3.3*	8.8	8.4	9.2	7.8	7.6	8.1	6.5	7.7	*5.7*	6.3	6.0	*5.6*
	CLM	*5.0*	8.3	7.8	8.5	8.3	*5.2*	*5.2*	5.8	9.3	6.0	8.9	6.0	6.7
	ALM	7.5	*3.6*	7.6	*4.5*	12.7	7.0	*4.7*	9.6	6.3	9.0	7.1	5.4	6.1
PSFM	SLM	*2.8*	9.4	7.9	9.8	6.0	6.5	8.9	7.5	8.2	6.7	5.9	*5.7*	*5.7*
	CLM	*3.8*	8.4	7.8	8.5	7.3	*3.8*	*5.1*	8.3	10.3	7.5	8.2	5.5	6.3
	ALM	5.3	*4.8*	7.9	5.3	12.5	5.4	*4.7*	10.3	7.7	9.8	6.4	*4.9*	5.9

Table 8 Top 10 combinations of similarity measures and linkage methods

Rank	1	2	3	4	5	6	7	8	9	10
PSFE	G1 (ALM)	G3 (ALM)	LIN (ALM)	VE (ALM)	VM (ALM)	MZ (ALM)	IOF (ALM)	SM (ALM)	G2 (ALM)	ES (ALM)
PSFM	LIN (ALM)	G1 (ALM)	VE (ALM)	G3 (ALM)	ES (ALM)	IOF (ALM)	VM (ALM)	SM (ALM)	ES (CLM)	IOF (CLM)

The next analysis deals with an influence of different linkage methods (link.), which are compared in Table 7. The ES measure created the best clusters over three linkages in spite of its slightly worse performance in the average linkage method. The newly introduced measures, VE and VM, were the least influenced by different linkages. The measures IOF and LIN perform better in complete and average linkages. Goodall's measures G1 and G3 prefer the average linkage method, whereas the measures SM and OF the single linkage method.

The last part of the experiment focuses mainly on a practical application of the examined similarity measures in hierarchical clustering, i.e., on determining which combinations of similarity measures and linkage methods produce the high-quality clusters. To achieve this goal, average PSFE and PSFM scores of 39 combinations of 13 similarity measures and three clustering methods were ordered and ranked in decreasing order. The results for the top ten combinations are presented in Table 8. It is apparent that the average linkage method outperforms the complete linkage and single linkage methods (in this order) in both the evaluation coefficients. The best clusters were produced as a combination of the G1 measure and ALM. They are followed by LIN with ALM, G3 with ALM, and VE with

Table 9 Top 10 combinations of similarity measures and linkage methods broken down by the number of variables

Index	var.	1	2	3	4	5	6	7	8	9	10
PSFE	4	G1 (ALM)	G3 (ALM)	VE (ALM)	LIN (ALM)	VM (ALM)	SM (ALM)	IOF (ALM)	MZ (ALM)	ES (ALM)	G2 (ALM)
	6	G1 (ALM)	LIN (ALM)	G3 (ALM)	VE (ALM)	VM (ALM)	IOF (ALM)	ES (ALM)	SM (ALM)	G2 (ALM)	MZ (ALM)
	8	G1 (ALM)	G3 (ALM)	LIN (ALM)	MZ (ALM)	VE (ALM)	VM (ALM)	SM (ALM)	IOF (ALM)	G2 (ALM)	ES (ALM)
	10	G1 (ALM)	LIN (ALM)	G3 (ALM)	MZ (ALM)	VE (ALM)	VM (ALM)	IOF (ALM)	G2 (ALM)	SM (ALM)	OF (CLM)
PSFM	4	VE (ALM)	IOF (ALM)	ES (ALM)	IOF (CLM)	ES (CLM)	LIN (ALM)	SM (ALM)	VM (ALM)	VE (CLM)	G1 (ALM)
	6	LIN (ALM)	ES (ALM)	IOF (ALM)	G1 (ALM)	VE (ALM)	VM (ALM)	G3 (ALM)	SM (ALM)	ES (CLM)	G2 (ALM)
	8	G1 (ALM)	G3 (ALM)	LIN (ALM)	VE (ALM)	ES (ALM)	VM (ALM)	IOF (ALM)	SM (ALM)	MZ (ALM)	G2 (ALM)
	10	LIN (ALM)	G1 (ALM)	G3 (ALM)	MZ (ALM)	VE (ALM)	VM (ALM)	IOF (ALM)	ES (ALM)	G2 (ALM)	SM (ALM)

Table 10 Recommended similarity measures based on the number of variables and the linkage method

Linkage	4–6 variables	8–10 variables
SLM	ES, VE	ES
CLM	ES, IOF	LIN, ES
ALM	VE, LIN	LIN, G1

ALM. The best similarity measure with the complete linkage was ES according to both the PSFE and PSFM coefficients, which was placed on the eleventh and ninth ranks respectively. Therefore, unless there is a specific reason for using the other linkage methods, the average linkage method should be preferred. The results for all 39 combinations occur in Tables 11 and 12 in the Appendix.

When examining the top ten combinations of similarity measures and linkage methods broken down by the number of variables in Table 9, the average linkage method dominates again. Altogether, they are three best-performing similarity measures with ALM there, namely LIN, G1, and VE. In accordance with the previous findings, the VE measure performs better by four variables in a dataset, whereas the LIN and G1 measures outperform it by a higher number of variables.

It is apparent that there does not exist one universal similarity measure. However, taking into account that the dataset difficulty does not play an important role in the clustering performance of the examined similarity measures, the main outputs of this experiment can be summarized in Table 10. Based on the number of variables and the used linkage methods, one or two suitable similarity measures (ordered by priority) are recommended.

6 Conclusion

In the paper, 13 similarity measures for categorical data were compared. Among them was the well-known simple matching coefficient, which served as a reference similarity measure, and also two newly proposed similarity measures, Variable Entropy (VE) and Variable Mutability (VM). The examined similarity measures were evaluated using the mean ranked scores of PSFE and PSFM evaluation coefficients on 60 generated datasets and with three commonly used linkage methods.

In the experiment, it was found out that a clustering performance of the examined similarity measures depends mostly on the number of variables and the used linkage method. The influence of the dataset difficulty, expressed by the numbers of categories, seemed to be negligible.

The examined similarity measures were evaluated first standalone and second in their combinations with linkage methods, which resembles their practical use. With respect to the first way, the best clusters were provided by the ES measure. The measure performed well, especially, in situations when it was used with the single or the complete linkage methods. The newly introduced similarity measures VE and VM provided very good clusters, especially, when the number of variables was up to six. The IOF measure can be recommended for all types of datasets if the complete linkage method is used. The LIN measure is strongly recommended to use in datasets with larger numbers of variables and the complete or the average linkage methods. The rest of the examined similarity measures cannot be recommended for a regular use in hierarchical clustering because their cluster quality is poor or unstable. The reference measures SM provided outputs around the average.

For the second way of evaluation, when the combinations of similarity measures and linkage methods were examined, the majority of similarity measures combined with the average linkage method outperformed their combinations with complete and single linkage methods, especially, with the LIN and G1 similarity measures. Therefore, if possible, the average linkage method with one of these similarity measures should be preferred.

Funding Information This paper was supported by the University of Economics, Prague under the IGA project no. F4/41/2016.

Appendix

Table 11 Rank orders of all combinations of similarity measures and linkage methods (PSFE)

1	2	3	4	5	6	7	8	9	10	11	12	13
G1	G3	LIN	VE	VM	MZ	IOF	SM	G2	ES	ES	IOF	LIN1
(ALM)	(ALM)	(ALM)	(ALM)	(ALM)	(ALM)	(ALM)	(ALM)	(ALM)	(ALM)	(CLM)	(CLM)	(ALM)
14	15	16	17	18	19	20	21	22	23	24	25	26
OF	LIN	VE	VM	LIN1	OF	G2	G1	SM	G3	MZ	G4	ES
(ALM)	(CLM)	(CLM)	(CLM)	(CLM)	(CLM)	(CLM)	(CLM)	(CLM)	(CLM)	(CLM)	(CLM)	(SLM)
27	28	29	30	31	32	33	34	35	36	37	38	39
VE	IOF	VM	G4	MZ	OF	SM	LIN1	LIN	G1	G3	G2	G4
(SLM)	(SLM)	(SLM)	(ALM)	(SLM)	(SLM)	(SLM)	(SLM)	(SLM)	(SLM)	(SLM)	(SLM)	(SLM)

Table 12 Rank orders of all combinations of similarity measures and linkage methods (PSFM)

1	2	3	4	5	6	7	8	9	10	11	12	13
LIN	G1	VE	G3	ES	IOF	VM	SM	ES	IOF	G2	MZ	VE
(ALM)	(ALM)	(ALM)	(ALM)	(ALM)	(ALM)	(ALM)	(ALM)	(CLM)	(CLM)	(ALM)	(ALM)	(CLM)
14	15	16	17	18	19	20	21	22	23	24	25	26
LIN	VM	OF	LIN1	SM	G4	ES	G2	G1	G3	OF	LIN1	MZ
(CLM)	(CLM)	(ALM)	(ALM)	(CLM)	(CLM)	(SLM)	(CLM)	(CLM)	(CLM)	(CLM)	(CLM)	(CLM)
27	28	29	30	31	32	33	34	35	36	37	38	39
IOF	VE	VM	G4	MZ	OF	SM	LIN1	LIN	G1	G3	G2	G4
(SLM)	(SLM)	(SLM)	(ALM)	(SLM)	(SLM)	(SLM)	(SLM)	(SLM)	(SLM)	(SLM)	(SLM)	(SLM)

References

Anderberg, M.R. (1973). *Cluster analysis for applications. Probability and mathematical statistics*. New York: Academic Press.

Boriah, S., Chandola, V., Kumar, V. (2008). Similarity measures for categorical data: a comparative evaluation. In *Proceedings of the eighth SIAM International Conference on Data Mining* (pp. 243–254).

Chandola, V., Boriah, S., Kumar, V. (2009). A framework for exploring categorical data. In *Proceedings of the ninth SIAM International Conference on Data Mining* (pp. 187–198): SIAM.

Chatuverdi, A., Foods, K., Green, P.E., Carroll, J.D. (2001). K-modes clustering. *Journal of Classification*, *18*(1), 35–55.

Chen, L., & Guo, G. (2014). Centroid-based classification of categorical data. In Li, F., Li, G., Hwang, S.-w., Yao, B., Zhang, Z. (Eds.) *Web-age information management* (pp. 472–475). Cham: Springer International Publishing.

Deza, M.M., & Deza, E. (2009). *Encyclopedia of distances*. Berlin: Springer.

Eskin, E., Arnold, A., Prerau, M., Portnoy, L., Stolfo, S. (2002). *A geometric framework for unsupervised anomaly detection*, (pp. 77–101). Boston: Springer US.

Everitt, B., Landau, S., Leese, M., Stahl, D. (2011). *Cluster analysis. Wiley series in probability and statistics*. New York: Wiley.

Goodall, D.W. (1966). A new similarity index based on probability. *Biometrics*, *22*(4), 882–907.

Gower, J.C. (1971). A general coefficient of similarity and some of its properties. *Biometrics*, *27*(4), 857–871.

Hennig, C., Meila, M., Murtagh, F., Rocci, R. (2015). Handbook of cluster analysis. Chapman & Hall/CRC Handbooks of modern statistical methods. Taylor & Francis.

Huang, Z. (1998). Extensions to the k-means algorithm for clustering large data sets with categorical values. *Data Mining and Knowledge Discovery*, *2*(3), 283–304.

Jaccard, P. (1912). The distribution of the flora in the alpine zone. *New Phytologist*, *11*(2), 37–50.

Lin, D. (1998). An information-theoretic definition of similarity. In *Proceedings of the 15th International Conference on Machine Learning* (pp. 296–304): Morgan Kaufmann.

Morlini, I., & Zani, S. (2012). A new class of weighted similarity indices using polytomous variables. *Journal of Classification*, *29*(2), 199–226.

Qiu, W., & Joe, H. (2015). clusterGeneration: random cluster generation (with specified degree of separation). R package version 1.3.4.

Qiu, W., & Joe, H. (2016). Generation of random clusters with specified degree of separation. *Journal of Classification*, *23*(2), 315–334.

Řezanková, H. (2009). Cluster analysis and categorical data. *Statistika*, *89*(2), 216–232.

Řezanková, H., Löster, T., Húsek, D. (2011). Evaluation of categorical data clustering. *Advances in Intelligent Web Mastering*, *3*, 173–182.

San, O.M., Huynh, V.N., Nakamori, Y. (2004). An alternative extension of the k-means algorithm for clustering categorical data. *International Journal of Applied Mathematics and Computer Science*, *14*(2), 241–247.

Shannon, C.E. (2001). A mathematical theory of communication. *ACM SIGMOBILE Mobile Computing and Communications Review*, *5*(1), 3–55.

Sokal, R.R., & Michener, C.D. (1958). A statistical method for evaluating systematic relationships. *University of Kansas Scientific Bulletin*, *28*, 1409–1438.

Spärck Jones, K. (1972). A statistical interpretation of term specificity and its application in retrieval. *Journal of Documentation*, *28*, 11–21.

Strauss, T., & von Maltitz, M.J. (2017). Generalising Ward's method for use with Manhattan distances. *PLoS ONE*, *12*(1), 1–21.

Šulc, Z., & Řezanková, H. (2015). nomclust: an R package for hierarchical clustering of objects characterized by nominal variables. In *Proceedings of the 9th International Days of Statistics and Economics* (pp. 1581–1590). Slaný: Melandrium.

Todeschini, R., Consonni, J., Xiang, H., Holliday, V., Buscema, M., Willett, P. (2012). Similarity coefficients for binary chemoinformatics data: overview and extended comparison using simulated and real data sets. *Journal of Chemical Information and Modeling*, *52*(11), 2884–2901.

Warrens, M.J. (2008). Similarity coefficients for binary data. Ph.D. thesis, University of Leiden.

Warrens, M.J. (2016). Inequalities between similarities for numerical data. *Journal of Classification*, *33*(2), 141–148.

Yi, J., Yang, G., Wan, J. (2016). Category discrimination based feature selection algorithm in chinese text classification. *Journal of Information Science and Engineering*, *32*(5), 1145–1159.

Yim, O., & Ramdeen, K.T. (2015). Hierarchical cluster analysis: comparison of three linkage measures and application to psychological data. *The Quantitative Methods for Psychology*, *11*(1), 8–21.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.